SAPTHAGIRI NPS UNIVERSITY
UNMATCHED EXCELLENCE, UNLIMITED POTENTIAL

Department of Computer Science & Engineering
Problem Solving Using C -25BEPHY104
AY-2025-26 Sem-I
Question Bank

MODULE-1				
Sl.No	**Question**	**RBT Level**	**COs**	**Marks**
1.	What is decision control? Why is it important in programming?	L2	CO2	10
2.	Differentiate between sequential, decision-making, and looping constructs.	L2	CO2	5
3.	Explain the flow of control in a decision-making process using a flowchart.	L2	CO2	5
4.	Describe the difference between compile-time and run-time decision control.	L3	CO2	7
5.	What is decision control? Why is it important in programming?	L3	CO2	5
6.	Differentiate between sequential, decision-making, and looping constructs.	L2	CO2	5
7.	Explain the flow of control in a decision-making process using a flowchart.	L2	CO2	10
8.	Explain the working of an **if statement** with syntax and example.	L3	CO2	5
9.	Differentiate between **if**, **if-else**, and **if-else-if ladder**.	L2	CO2	10
10.	Write a program to find the largest of three numbers using nested **if** statements.	L3	CO2	5
11.	What is a **switch statement**? How is it different from an if-else ladder?	L2	CO2	5
12.	Write a program using a **switch statement** to perform arithmetic operations based on user choice.	L3	CO2	10
13.	What happens if we omit the **break** statement in a switch-case construct?	L2	CO2	10
14.	What are iterative statements? List the different types of loops.	L2	CO2	5
15.	Explain the **for loop** with syntax and example.	L2	CO2	5
16.	How does a **while loop** differ from a **do-while loop**?	L2	CO2	5
17.	Write a program to print all even numbers between 1 and 50 using a **while loop**.	L2	CO2	10
18.	Write a program to calculate the factorial of a number using a **for loop**.	L2	CO2	10
20.	What are the advantages of using loops in programming?	L2	CO2	10

22.	Explain how nesting of different types of loops (for, while) can be done.	L2	CO2	10
23.	What is the purpose of the **break** statement? Give an example.	L2	CO2	10
24.	What is the difference between **break** and **continue** statements?	L2	CO2	10
25.	Write a program to print numbers from 1 to 10, skipping 5 using **continue**.	L2	CO2	10
26.	Explain the **goto** statement with syntax and example.	L2	CO2	10
27.	Why is the use of **goto** discouraged in modern programming?	L2	CO2	10
28.	Write a program to demonstrate the use of **break** inside nested loops.	L2	CO2	10
29.	Write a program to check whether a number is positive, negative, or zero using if-else.	L3	CO2	10
30.	Write a program to display the multiplication table of a number using loops.	L3	CO2	10
31.	Write a program to find the sum of digits of a number using a while loop.	L3	CO2	10
32.	Write a program to print Fibonacci series up to `n` terms using loops.	L3	CO2	10
33..	Write a program to check whether a given number is a palindrome.	L3	CO2	10
34.	Write a program to find the reverse of a given integer using loop constructs.	L3	CO2	10
35.	Write a menu-driven program using a switch-case for basic calculator operations.	L3	CO2	10